



PRODUCT NAME

High-performance Dishwashing Liquid Formula

Description

Manual dishwashing detergent for kitchen use.

Technical Specifications

pH Level:	7
Viscosity:	Medium
Shelf Life:	24 months
Batch Size:	100-500 kg
Processing Time:	2-4 hours
Temperature:	Room temperature
Storage Conditions:	Cool, dry place
Equipment:	Standard mixer
Certification:	Meets industry standards

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Sodium Laureth Sulfate	Sodium Laureth Sulfate	15.31%	Primary surfactant for cleaning and foaming.
2	Cocamidopropyl Betaine	Cocamidopropyl Betaine	10.2%	Secondary surfactant for mildness and foam stabilization.
3	Sodium Citrate	Sodium Citrate	5.1%	Builder for enhanced cleaning performance.
4	Water	Aqua	65.31%	Solvent and carrier for ingredients.
5	Sodium Chloride	Sodium Chloride	1.53%	Viscosity modifier.
6	Fragrance	Fragrance	0.51%	Provides pleasant scent to the product.
7	Preservative (such as Sodium Benzoate)	Sodium Benzoate	0.51%	Prevents microbial growth and extends shelf life.
8	Citric Acid	Citric Acid	1.53%	pH adjuster to maintain pH at 7.
Total			100%	

Manufacturing Process

Phase 1: Manufacturing Phase

Phase 2: Manufacturing Phase

Phase 3: Manufacturing Phase

Phase 4: Manufacturing Phase

Phase 5: Manufacturing Phase

Phase 6: Manufacturing Phase

Phase 7: Manufacturing Phase

Phase 8: Manufacturing Phase

Phase 9: Manufacturing Phase

Required Equipment

Standard mixer

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Cool, dry place

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: Meets industry standards

Scaling Note

- Lab Scale:** This formulation is designed for laboratory testing and development.
- Pilot Scale:** For pilot production, scale proportionally and verify all parameters.
- Production Scale:** Consider equipment limitations, mixing efficiency, and process validation.
- Batch Size:** Current formulation is optimized for 100-500 kg.
- Scaling Factor:** Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply as needed according to product instructions